DEB portal

Zero-variate data

Zero-variate data is a scalar (so a single number) that quantifies a feature of the organism at a given moment. Auxiliary data might be associated to any data, so also to zero-variate data. Although a single data-point has little information for parameter estimation in itself, a set of different data-points can be very useful, certainly in combination with uni- and bi-variate data sets. A prediction for a zero-variate data set is typically a function of parameter values, which should be seen as removing one degree of freedom in the parameter estimation. So with 10 parameters, 10 (independent) zero-variate data sets theoretically determine all 10 parameters.

Codification of types zero-variate data

In the mydata file the author needs to specify which types of data are used in the entry. Labels for zero-variate data are packed as metaData.data_0 in the mydata-file. The codification generally follows DEB's notation rules. The leading character indicates the type of variable, followed by a marker for the life history event.

Indices symbolising different life history events

0 start of development	j end of acceleration	R first breeding
h hatching (leaving egg)	p puberty (start of allocation to reproduction)	e emergence (imago)
b birth (start of feeding)	i ultimate state	s start of acceleration
x weaning (or fledging)	m death (life span)	

Optionally this can be followed by an underscore ("_"), indicating several values of a modifying factor:

f functional response	T temperature	t time
L length	W weight	

The labels are stylized, meaning e.g. that label ap stands for age-at-puberty, while actually time since birth at puberty, tp, might be used as a data-point. The rationale is that delays in the start of development can occur, which would affect ap, but not tp. Data might relate to males and females; this is not indicated in the data labels, but comments in such entries specify the parameters in which they differ.

Data labels	Description
ah	age at h
ah_T	age at h (several T))
ab	age at birth
ab_T	age at birth (several T)
ab_f	age at birth (several f)
tg	gestation time
ax	age at x
as	age at s
aj	age at j
aj_T	age at j (several T)
ap	age (or time since birth) at p
ap_T	age at p (several T)
ap_f	age at p (several f)
aR	age at R
ae	age at e
astar	instar duration
am	age at death (life span)
am_f	age at death (several f)
am_T	age at death (several T)
L0	length at 0
Lh	length at h
Lb	length at b
Lb_f	length at b (several f)
Lx	length at x
Ls	length at s
Lj	length at j
Lp	length at p
Lp_f	length at p (several f)
LR	length at R
Lstar	length at instar
Li	length at i
Li_f	length at i (several f)
L_t	length at time t
L_W	length at time weight

dLb	change in length at b
dLp	change in length at p
dLj	change in length at j
Ab	surface area at b
Ap	surface area at p
Ai	surface area at i
V0	volume at 0
Vb	volume at b
Vb_f	volume at b (several f)
Vp	volume at p
Vp_f	volume at p (several f)
Vi	volume at i
Vi_f	volume at i (several f)
Ww0	wet weight at 0
Wwh	wet weight at h
Wwb	wet weight at b
Wwb_f	wet weight at b (several f)
Wwx	wet weight at x
Wws	wet weight at s
Wwj	wet weight at j
WwR	wet weight at R
Wwp	wet weight at p
Wwe	wet weight at e
Wwi	wet weight at i
Ww_L	wet weight at lenght
Ww_t	wet weight at time
dWw_W	change in wet weight at W
Wd0	dry weight at 0
Wdh	dry weight at h
Wdb	dry weight at b
Wdb_f	dry weight at b (several f)
Wdx	dry weight at x
Wds	dry weight at s
Wdj	dry weight at j
Wdp	dry weight at p
Wde	dry weight at e
Wdi	dry weight at i
Wd_t	dry weight at t
Wd_L	dry weight at length
WdR_L	gonad dry weight at L
WdR_Wd	gonad dry weight at Wd
WC0	carbon weight at 0
WN0	nitrogen weight at 0
dWdb	change in dry weight at b
dWd	decrease in dry weight during starvation
E_L	total energy at L
E0	reserve energy at 0
Eh	reserve energy at h
Eb	reserve energy at b
Ej	reserve energy at j
EbE0	ratio of reserve energy at b and 0
EXx	cumulated food energy at x
r	specific growth rate
rB	von Bertalanffy growth rate
rB_T	von Bertalanffy growth rate (several T)
Ri	reproduction rate at i
Ri_f	reproduction rate at i (several f)
R_L	reproduction rate at length
R_W	reproduction rate at weight
GSI	gonado-somatic index
Ni	(total) number of offspring at i
Ni_f	(total) number of offspring at i (several f)
Fm	maximum clearance (filtering) rate
K	half saturation coefficient
pXi	maximum ingestion rate (energy)
pXi_Wd	maximum ingestion rate at Wd (energy)
pX_t	food energy ingestion rate at t
pAi	maximum assimilation rate (energy)
pL	lactation flux (energy)
kapX	digestion efficiency
RQ	respiration quotient: JC/JO
JXi	food consumption at i

JX_L	food consumption at L
JX_W	food consumption at W
JLi	milk production at i
JOb	O2 consumption at b
JOi	O2 consumption at i
JO_W	O2 consumption at W
JCi	CO2 production at i
ss	supply stress
xi_WE	energy density of dry mass excluding the reproduction buffer